Robin **Wroblowski**

Ph.D. Candidate in Neuromorphic Control & Spiking Neural Circuits

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Education

KU Leuven — Leuven, Belgium

Ph.D. in Neuromorphic Control — 2023 - Present

- Neuromorphic Systems Lab

KU Leuven — Leuven, Belgium

Master in Mathematical Engineering — *2021 – 2023*

- Graduated *cum laude*
- Exchange Semester at the Technical University of Vienna (Erasmus+, 2021)

Bachelor in Engineering Science — *2018 – 2021*

- Computer Science & Electrical Engineering
- Graduated *cum fructu*

Doctoral Research & Teaching

Ph.D. Candidate in Neuromorphic Control — KU Leuven

Advised by Prof. Rodolphe Sepulchre — 2023 - Present

- **Design of Pattern Generating Networks:** Proposed a shift from coupled-oscillator networks to excitable networks based on event-propagation to achieve a robust framework for distributed design and analysis of Central Pattern Generators.
- **Regulating Rhythmic Interaction:** Developed methods for controlling the behavior of the event-based interconnection between a rhythmic plant and a neuromorphic controller by balancing the relative roles of feedforward and feedback.
- **Energy-based Modeling of Excitability:** Constructed an energy-based formulation of neuronal excitability to enable energetically consistent, domain-agnostic realizations of excitable networks (e.g., fluidic CPGs).

Teaching Assistant — KU Leuven

Faculty of Engineering Science — 2023 - Present

- Instructed and prepared materials for a Bachelor-level Systems & Control course over four years (serving as Lead Teaching Assistant for two years). Coordinated exercise sessions, designed exam questions, and managed grading.
- Provided daily supervision and mentorship to a Master's student for their thesis, "Neuromorphic Control of a Pendulum," demonstrating the practical verification of rhythmic neuromorphic control principles using a physical pendulum.

Publications

Describing functions and phase response curves of excitable systems — European Control Conf.

Robin Wroblowski, Rodolphe Sepulchre — 2026

- Published in the 2026 European Control Conference (ECC).

Neuromorphic design of a rhythmic network — arXiv Preprint

Robin Wroblowski, Rodolphe Sepulchre — 2026

- Available as arXiv preprint: arXiv:2610.39762.
- Submitted to Journal of Neuromorphic Computing & Engineering.

On energy-based modelling for fluidic excitable neurons — Under Review

Robin Wroblowski, Brent Michiels, Benjamin Gorissen, Rodolphe Sepulchre — Submitted 2026

- Submitted to the 2027 European Control Conference (ECC).

Regulating event-based interaction through rhythmic compliance — In Preparation

Robin Wroblowski, Rodolphe Sepulchre — Expected January 2027

- Target submission: January 2027, Automatica / Journal of Neuromorphic Computing and Engineering

CPG design for adaptive soft robotic locomotion and reflexive interaction — In Preparation

Robin Wroblowski, Brent Michiels, Benjamin Gorissen, Rodolphe Sepulchre — Expected April 2027

- Manuscript in preparation for submission in April 2027.

Other Experience

Vice-President Leuven, Belgium

Vlaamse Technische Kring (VTK) 2022 - 2023

- Directed internal relations and long-term strategy for the official engineering student association of 84 members representing over 4,000 students, and managed the daily operations across multiple organizational subdivisions.

IT Team Leader Leuven, Belgium

Vlaamse Technische Kring (VTK) 2020 - 2022

- Led the development and deployment of multiple software projects, including a large-scale ticketing service, a career profile matching platform and a logistics backend to support campus-wide events.

Skills

Scientific Computing	**Python**, **Matlab**, **Fortran**, Julia, Magma
Software/Hardware	**C++**, **Arduino**, **C**, PHP, Java, JavaScript
Spoken Languages	**Dutch (Native)**, **English (Native)**, **German (Advanced)**, French (Intermediate)

Presentations & Talks

Describing Functions and Phase Response Curves of Excitable Systems

European Control Conference (ECC) July 2026

Regulating Rhythmic Compliance in Central Pattern Generators

45th Benelux Meeting on Systems and Control March 2026

Rebound Amplifiers for Discrete-Event Pattern Generation

44th Benelux Meeting on Systems and Control March 2025

Tuning the Entrainability of a Half-Center Oscillator

43rd Benelux Meeting on Systems and Control March 2024

References

Prof. Rodolphe Sepulchre
PhD Advisor
Department of Electrical Engineering (STADIUS)
KU Leuven, Belgium
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Prof. Fulvio Forni
Research Host & Collaborator
Department of Engineering
University of Cambridge, United Kingdom
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Prof. Benjamin Gorissen
PhD Co-Supervisor
Department of Mechanical Engineering (Soft Robotics Group)
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Prof. Daan Huybrechs
Master's Thesis Supervisor
Department of Computer Science
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